

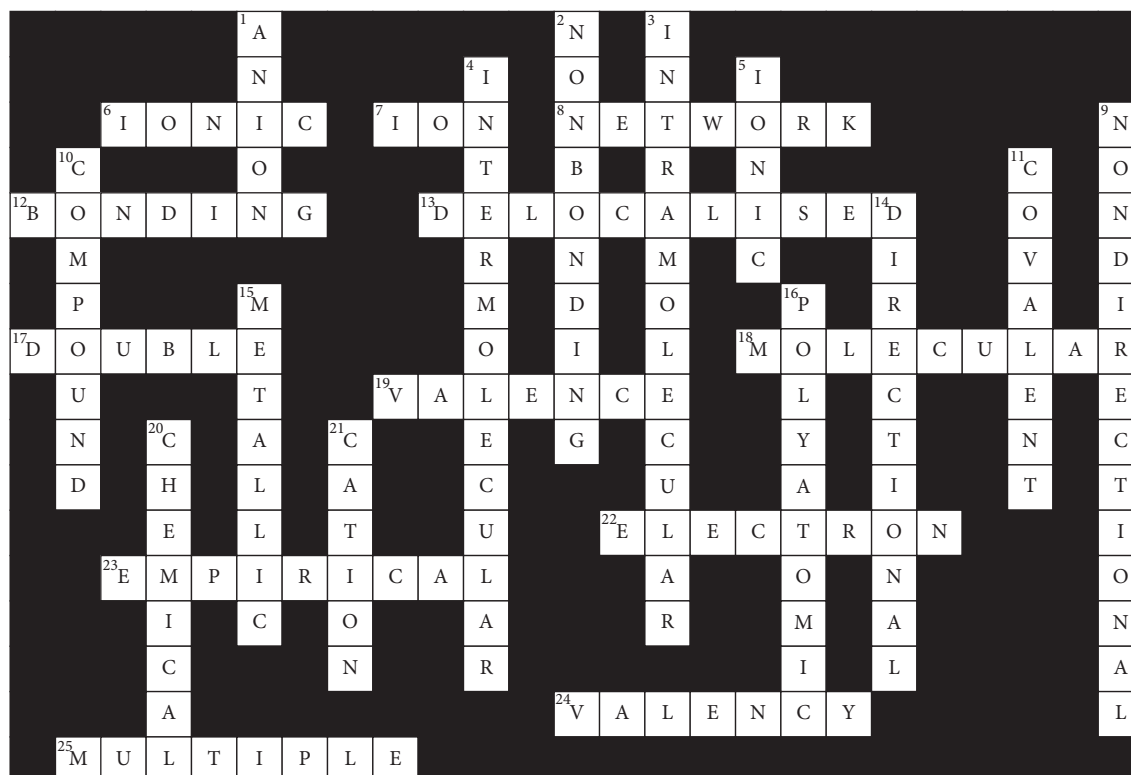
ACTIVITY SHEET ANSWERS**Chapter 3 Suggested answers****Prior knowledge**

- 1 Having equal numbers of negatively charged electrons and positively charged protons.
- 2 No, most types of electrically neutral are not chemically stable.
- 3 Having a full outer electron shell (valence shell)
- 4 Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe) and Radon (Rn) (All of which are found in Group 18 of the periodic table.)
- 5 By either taking electrons from, losing electrons to or sharing electrons with other atoms, so that they now have full outer shells
- 6 An element is a pure substance made of only one type of atom.
- 7 Non-metal atoms, because they have nearly full outer shells and can readily pull electrons off other metal atoms
- 8 Metal atoms, because they have a few electrons in their outer shells that are weakly held and therefore readily lost to non-metal atoms
- 9 Ions are charged atoms, which have unequal numbers of negatively charged electrons and positively charged protons.
- 10 They form negative ions because when they take electrons off metal atoms they now have excess negative charge.
- 11 They form positive ions because when they lose electrons to non-metal atoms they now have excess positive charge.
- 12 The oppositely charged metal and non-metal ions are electrically attracted to each other, forming ionic bonds.
- 13 Ionic bonding causes the ions to become arranged so that each positive ion is surrounded by negative ions and vice versa in a regular way.
- 14 A compound is a pure substance consisting of different types of atoms bonded together. An ionic compound is held together by ionic bonds.
- 15 A few loosely held outer shell electrons of each atom wander freely throughout the metal.
- 16 The attraction between the free electrons and the positively charged nuclei of surrounding atoms results in metallic bonding.
- 17 High melting points, good electrical and heat conductivity, ductility and malleability, metallic lustre etc.
- 18 Electrons are shared in pairs by two non-metal atoms, with each atom contributing one electron.
- 19 Both identical and non-identical non-metal atoms can share electrons.

20 The shared electrons are electrically attracted to the positive nuclei of both of the two atoms involved, resulting in a covalent bond.

21 Molecular compounds (molecules) and network solid compounds

Activity sheet 3.1

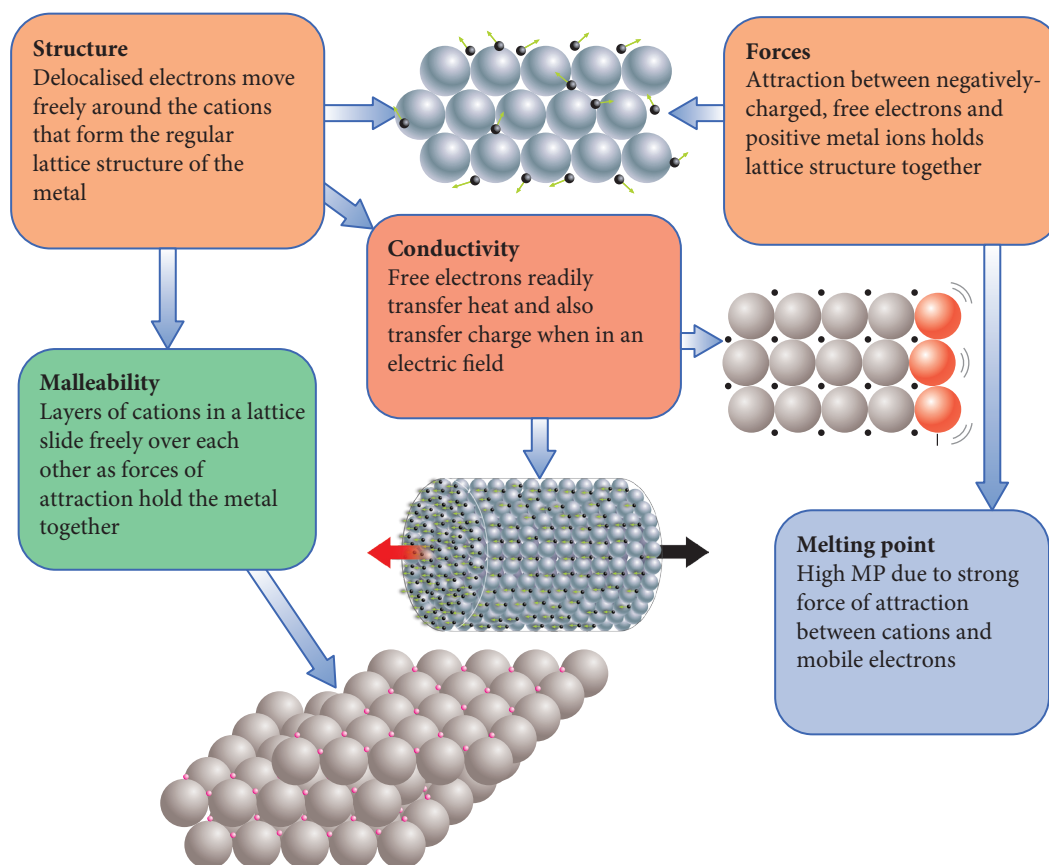


Activity sheet 3.2

Part A

- 1 The attraction between the negative and positive ions is such that each negative ion gets surrounded by positive ions and vice versa.
- 2 Attraction occurs between the oppositely-charged ions and repulsion between like-charged ions.
- 3 In the molten state the ions become free to travel and thus can transfer charge.
- 4 Like-charged ions are forced next to each other and the repulsion shatters the crystal.

Part B



- 1 The attraction between the metal cations and the sea of delocalised electrons flowing around them.
- 2 Loosely-held valence electrons roam freely, resulting in positively charged metal atoms (cations).
- 3 Non-directional forces of attraction occur between the metal cations and the mobile electrons.
- 4 The layers of the lattice slide easily over each other while the mobile electrons keep moving freely ensuring that the attraction between cations and them holds the metal together.

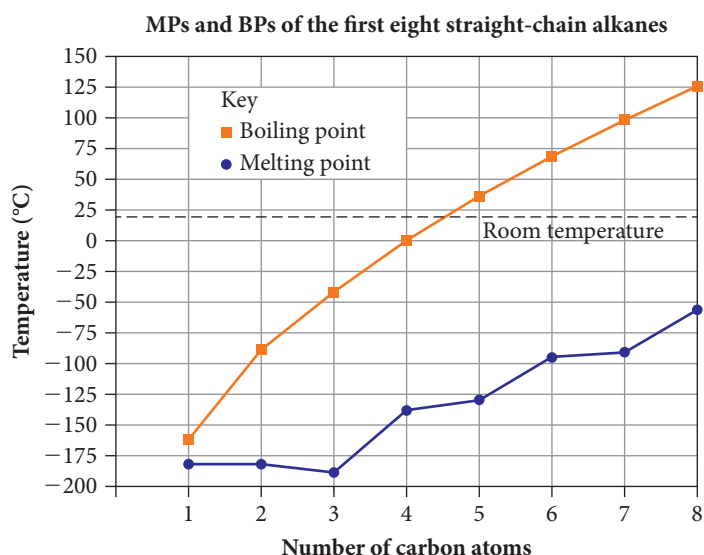
Activity sheet 3.3

Properties	Salt	Graphite	Aluminium
Melting point	Very high	Very high	Very high
Boiling point	Not relevant	Not relevant	Not relevant
Thermal conductivity	Very poor	Moderate	Very good
Electrical conductivity	Non-conductor	Very good	Excellent
Hardness	Hard but brittle	Quite soft	Quite hard
Malleability	Not malleable	Not malleable	Very malleable
Ductility	Not ductile	Not ductile	Very ductile
Solubility in water	Very soluble	Insoluble	Insoluble
Type of bonding	Ionic	Covalent network	Metallic

Properties	Sugar	Glass	Polythene
Melting point	Fairly low	Very high	Fairly low
Boiling point	Not relevant	Not relevant	Not relevant
Thermal conductivity	Not relevant	Very poor	Poor
Electrical conductivity	Not relevant	Good insulator	Insulator
Hardness	Easily crushed	Very hard	Soft
Malleability	Not relevant	Not malleable	Not relevant
Ductility	Not relevant	Not ductile	Not relevant
Solubility in water	Soluble	Insoluble	Insoluble
Type of bonding	Covalent molecular	Covalent network	Covalent molecular

Properties	Butter	Washing soda	Copper
Melting point	Very low	Decomposes at 50°C	Very high
Boiling point	Not relevant	Not relevant	Not relevant
Thermal conductivity	Not relevant	Not relevant	Excellent
Electrical conductivity	not relevant	Not relevant	Excellent
Hardness	Very soft	Hard but brittle	Hard
Malleability	Not relevant	Not relevant	Very malleable
Ductility	Not relevant	Not relevant	Very ductile
Solubility in water	Insoluble	Soluble	Insoluble
Type of bonding	Covalent molecular	Ionic	Metallic

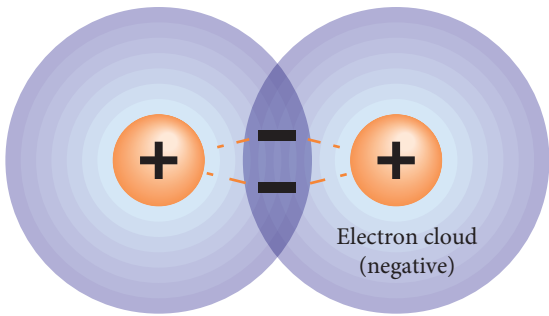
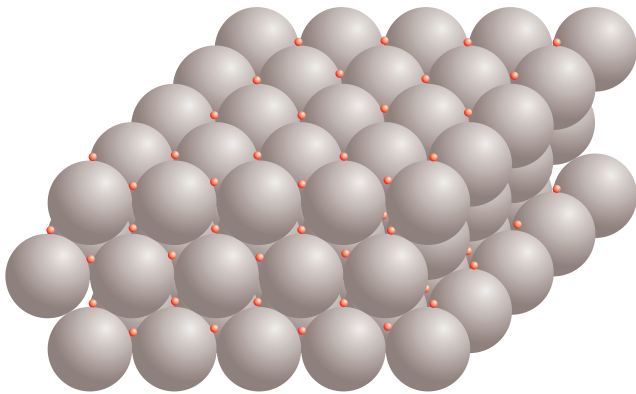
Activity sheet 3.4

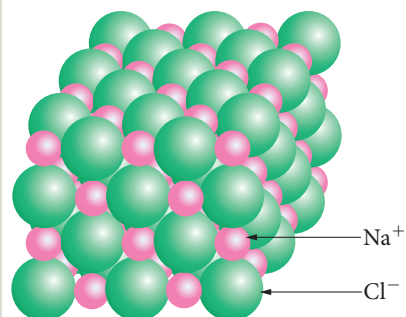
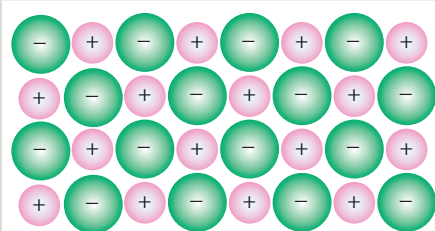


- 1
 - a Methane, ethane, propane and butane will be gases at room temperature.
 - b Because their boiling points are all below 20°C
- 2
 - a Pentane, hexane, heptane and octane will be liquids at room temperature.
 - b Because their melting points are all below 20°C and their boiling points are all above 20°C
- 3
 - a None of them.
 - b Because all of the alkanes on the graph have melting points below 20°C

- 4 The melting points decrease initially then rise but in a flattened zig-zag pattern as the number of carbon atoms in the alkane increases.
- 5 The boiling points increase regularly as the number of carbon atoms in the alkane increases.
- 6 The increasing size of the alkanes would result in stronger attraction between molecules.
- 7 About -40°C
- 8 About 160°C

Activity sheet 3.5

Bonding models	Distinctive features	Common features
 Covalent molecular	Only occurs between two non-metal atoms and bonding is directional Electrons are shared in pairs, one contributed by each atom Both achieve a more stable energy state by having a fuller valence shell The attraction that the two nuclei have for the shared electrons holds the molecule together	Atoms form bonds in order to attain a more stable energy state by having a full valence shell. Chemical bonding only involves valence electrons. All bonding involves the electrostatic force of attraction between positive and negative charges in participating atoms. Electrons experience a force of attraction from more than one nuclei. Only the outer regions of bonded atoms are in contact in chemical bonding.
 Key:  Sodium ion  Mobile valence electron Metallic bonding	Only occurs in metals loosely held valence electrons become delocalised Metal atoms become cations arranged in a lattice structure Mobile electrons move freely throughout structure Non-directional attraction between cations and mobile electrons holds the lattice structure together	



Ionic bonding

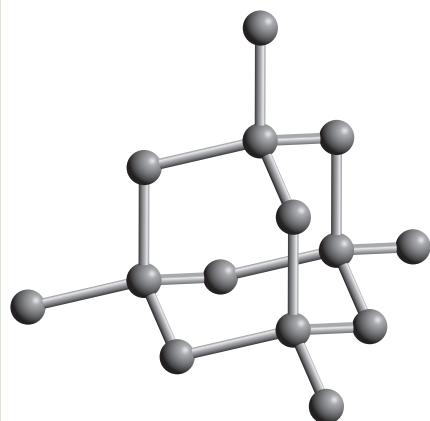
Involves the transfer of electrons from metal to non-metals atoms

Metal atoms become cations, non-metal atoms become anions

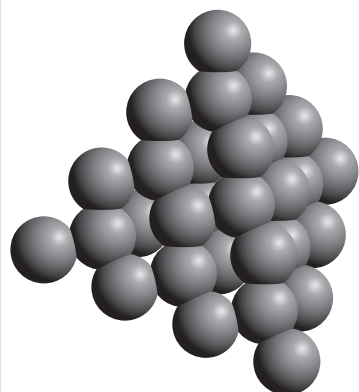
Oppositely charged ions are attracted to each other, but the bonding is non-directional

Minimal energy stable lattice arrangement has each cation surrounded by anions and vice versa

The ball-and-stick model



The 'solid ball' model



Covalent network

Only occurs between non-metal atoms, either the same or different

Each atom is covalently bonded to more than one other atom, forming a three-dimensional network

Networks extend indefinitely

Three dimensional network holds atoms rigidly in position

Bonding is directional

Chapter 3 Revision

- 1 a $\rightarrow$ Q, b $\rightarrow$ R, c $\rightarrow$ T, d $\rightarrow$ C, e $\rightarrow$ U, f $\rightarrow$ D, g $\rightarrow$ L, h $\rightarrow$ W, i $\rightarrow$ J, j $\rightarrow$ K, k $\rightarrow$ H, l $\rightarrow$ O, m $\rightarrow$ N, n $\rightarrow$ F, o $\rightarrow$ A, p $\rightarrow$ G, q $\rightarrow$ B, r $\rightarrow$ I, s $\rightarrow$ S, t $\rightarrow$ E, u $\rightarrow$ P, v $\rightarrow$ M, w $\rightarrow$ V
- 2 a *Anions* are negatively-charged atoms or groups of atoms. *Cations* are positively-charged atoms or groups of atoms
- b *Ionic bonding* is the electrostatic attraction between positively-charged cations and negatively-charged anions. *Covalent bonding* is the electrostatic attraction between shared negatively-charged electrons and positively-charged nuclei. *Metallic bonding* is the electrostatic attraction between negatively-charged delocalised valence electrons and positively-charged metal cations.
- c *Covalent molecular compounds* consist of discrete molecules in which the atoms are held together by covalent bonding. *Network covalent compounds* consist of a three-dimensional network of atoms held together by covalent bonding.
- d *Directional bonding* is in a direct line between adjacent atoms and it is typical of covalent bonding. *Non-directional bonding* occurs in most directions and it is typical of metallic and ionic bonding.
- e *Intermolecular bonds* are the electrostatic forces of attraction between molecules in close proximity. *Intramolecular bonds* are the electrostatic forces of attraction that hold atoms together within a molecule.
- 3 Metallic bonding is the electrostatic attraction between the 'sea' of delocalised valence electrons wandering through the metal and the positively-charged ions that result when the electrons detach from the atoms.

Ionic bonding is the electrostatic attraction between oppositely-charged ions, which are arranged in a regular array called a lattice. Each negatively-charged ion is surrounded by positively-charged ions and vice versa.

Molecular covalent bonding is the electrostatic attraction that occurs between the positively-charged nuclei and the pairs of electrons that are being shared between two atoms in a molecule.

Network covalent bonding involves atoms that are covalently bonded to more than one other atom, forming a three-dimensional network that extends indefinitely.
- 4 a Metals are good conductors of electricity because the delocalised valence electrons are free to transport charge in response to an electric field. These electrons drift towards the positive terminal of a circuit.
- b Metals are malleable and ductile because when the layers of positively charged ions slide over one another, the delocalised electrons rapidly adjust without disrupting the strength of the metallic bonding.
- 5 a Ionic compounds have high melting points because of the strong electrostatic attraction between oppositely-charged ions. A lot of energy is needed to free up the ions in order to form a liquid, which results in the high melting points.
- b Ionic compounds are hard because the ions are held rigidly in place in a lattice, but when strong shear forces act on the compound, ions of the same charge are pushed close together thus shattering the crystal.

- 6 a Molecular compounds have low boiling points because the forces that bind the molecules together (intermolecular forces) in the liquid or solid state are weak. This is despite the fact that the electrostatic forces holding the atoms in each molecule together (intramolecular forces) are strong.
- b Molecular compounds do not conduct electricity as there are no charged particles that are free to move.
- 7 a Covalent network compounds have very high melting points because the three-dimensional network of bonds means the atoms are all held rigidly in position. This means a lot of energy is required order to melt the solid, resulting in a high melting point
- b Graphite is a very good conductor of electricity because only three of each carbon atom's four valence electrons are involved in covalent bonds, which leaves delocalised electrons that are free to move within each layer. These delocalised electrons can conduct electricity.
- 8 a 1
- b 2
- c 3
- d 3
- e 0
- f 1
- g 2
- h 1
- i 4
- j 3
- k 0
- l 1

9

Bonding				
Property	Covalent molecular	Covalent network	Ionic	Metallic
Electrical conductivity	Does not conduct	Do not conduct (except for graphite)	Solids don't conduct but molten and aqueous states do	Good conductor in solid and liquid states
Hardness and malleability	Mostly soft	Hard and brittle	Hard and brittle	Hard and malleable
Melting and boiling points	Low	High	High	Variable but most commonly high
Solubility	Depends on solvent	Insoluble	Generally soluble in water	Generally insoluble

10 a ionic

- b** covalent network
- c** metallic
- d** covalent molecular
- e** metallic
- f** covalent molecular
- g** ionic
- h** covalent molecular

11 a Covalent molecular bonding, because of their low melting and boiling points, softness, and non-conductivity

- b** Weak, because they have low melting and boiling points
- c** Strong, because of the covalent bonds that hold the atoms together in each molecule

12 An addition reaction is when an additional atom is added to a molecule. Alkenes are unsaturated molecules, which means they have a double bond. The double bond can be broken allowing an extra hydrogen atom to be added one of the carbon atoms. Alkanes are saturated molecules, not having any double bonds, which means that no additional atoms can be added to the molecule.